



H2S

BHARATIYA ANTARIKSH HACKATHON 2026



Team Name : **SuryaSetu**

Team Leader Name : **Harsh Kawatra**

Problem Statement : **PS-15 | Nowcasting and Forecasting of Solar Flares from Aditya-L1 X-ray data**

SoLEXS 2-22 keV HELIOS 8-150 keV LightGBM + MiniROCKET TSS-driven Zero cost on a laptop

Team Members

Team Leader

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Team Member-2

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Team Member-3

Name: -

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A two-person, two-institute team: **full-stack and AI/ML engineering** applied to ISRO's own solar mission.

Opportunity and USP: why SuryaSetu wins

Most flare tools use a single channel from Earth orbit, and most "forecasts" are really just detections after the flare has already begun. **SuryaSetu is built differently on three axes:**

1. Dual-payload fusion.

SoLEXS measures **soft X-rays** (the heat a flare gives off) and HEL10S measures **hard X-rays** (the instant energy is released).

We merge both onto a **single timeline** from the L1 vantage, so the model reads the **full physics** of a flare instead of half of it - *the biggest reason our forecasts beat single-channel tools.*

2. Physics-grounded lead time.

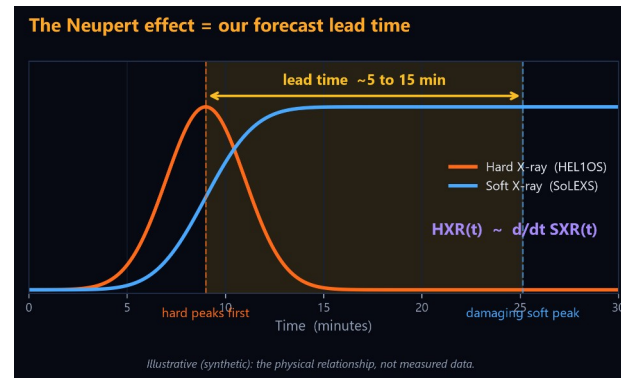
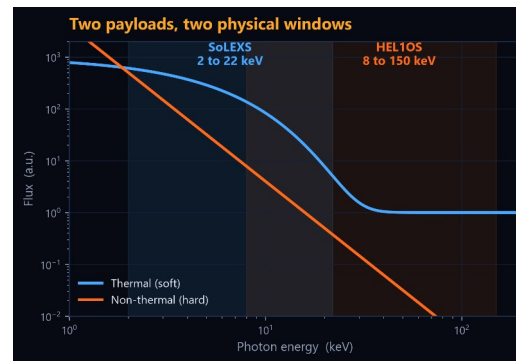
Hard X-rays surge several minutes before the damaging **soft-X-ray peak**, a real published relationship called the **Neupert effect**.

We convert that natural head-start into the actual **warning time** an operator needs, so the forecast rests on solar physics, *not a curve fit that only matches past data.*

3. Honest, operational evaluation.

Flares are rare, so a model that always says "no flare" looks **99% accurate** and is useless.

We score with the **True Skill Statistic**, calibrate every probability, and validate only on **future time blocks** - the standard ISRO and the space-weather community actually trust.



USP: the only dual-payload, L1-vantage system that turns the Neupert effect into calibrated, TSS-validated lead-time warnings, reproducible at zero cost on a laptop.

Features offered by the solution

01

Automated nowcast catalogue

Real-time soft and hard detection merged into one master catalogue: onset, peak, end, class and SNR.

02

Calibrated forecast + lead time

P(flare within N minutes) with calibrated confidence and a measured lead-time distribution.

03

Live dashboard + visual alerts

Dual light curves, hard/soft ratio and a green, amber, red banner that flips before the peak.

04

Full A-to-X dynamic range

Adaptive baselining plus dual SoLEXS detectors catch faint A/B/C and giant M/X flares.

05

Explainable alerts (SHAP)

Every forecast states which precursor fired it, so scientists can trust and audit it.

06

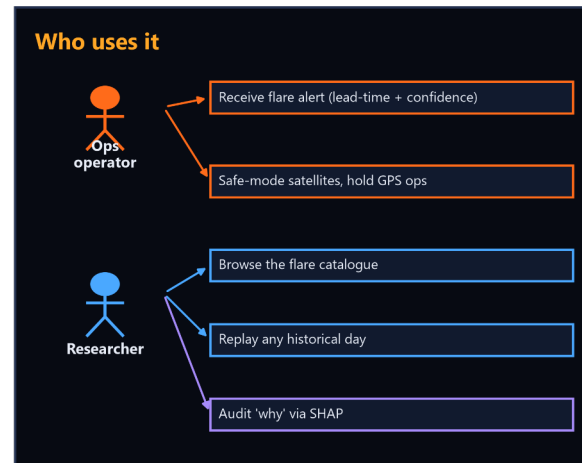
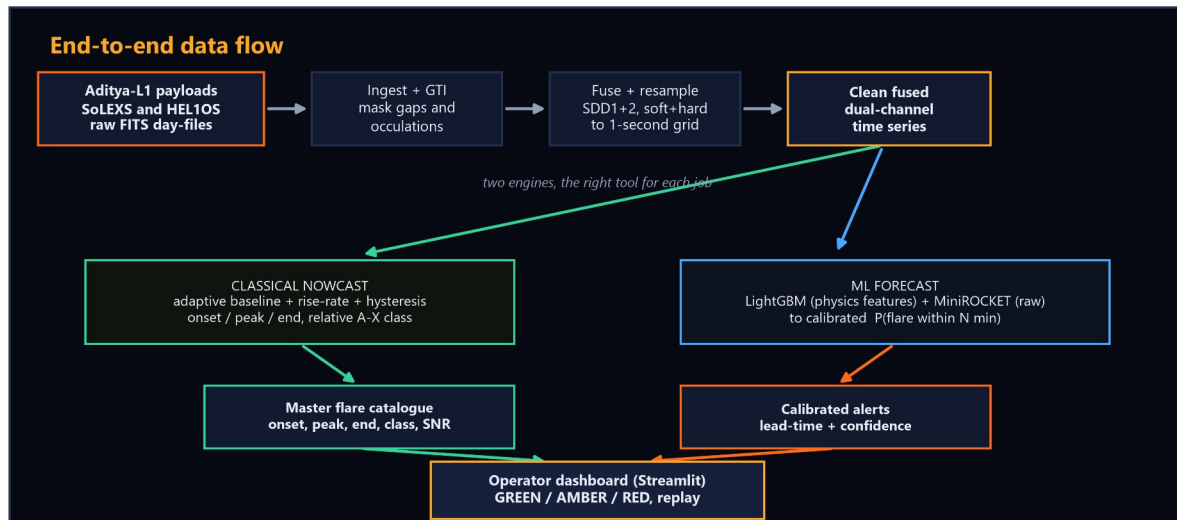
Offline and reproducible

No cloud, no cost, no proprietary software: end-to-end on one laptop from public data.

How it works in practice: the system reads both Aditya-L1 X-ray channels continuously, **automatically logs every flare** it detects into the catalogue, and **forecasts the next flare's strength and timing** with a **calibrated confidence**.

The operator watches one dashboard; when risk crosses a tuned threshold the **banner turns red** with the **minutes of lead time** remaining, and **SHAP** names the exact precursor that triggered it.

Process flow and use-case



In plain terms: raw satellite files are **cleaned** (bad seconds removed) and merged into one **1-second timeline**.

That stream splits into **two engines** running side by side: a **classical detector** that logs flares the moment they start, and an **ML forecaster** that predicts the next one.

Both feed the **catalogue, the alerts, and the live dashboard**.

Two users.

An **operations engineer** receives an alert and safe-modes satellites or holds GPS-critical work;

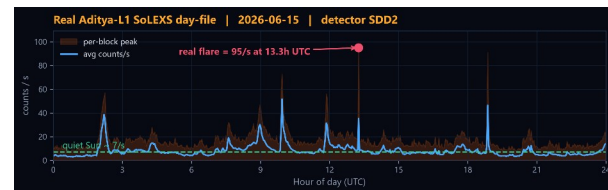
a **researcher** browses the catalogue, replays any past day, and audits why each alert fired.

Wireframe: the operator dashboard



One screen for the operator: the live soft and hard light curves with a **moving playhead**, the rising **hard/soft ratio** that acts as the early warning, a **probability gauge** for the next N minutes, and the running **flare catalogue**.

The banner **flips red** before the **soft-X-ray peak**, and *that gap is the warning time we deliver*.



Renders real Aditya-L1 data (2026-06-15)

QUIET

The Sun is **calm** and **no action is needed**.

The system keeps logging the **quiet-Sun baseline** it measures against.

WATCH

The **hard/soft ratio is climbing**, the physics precursor that often comes before a flare.

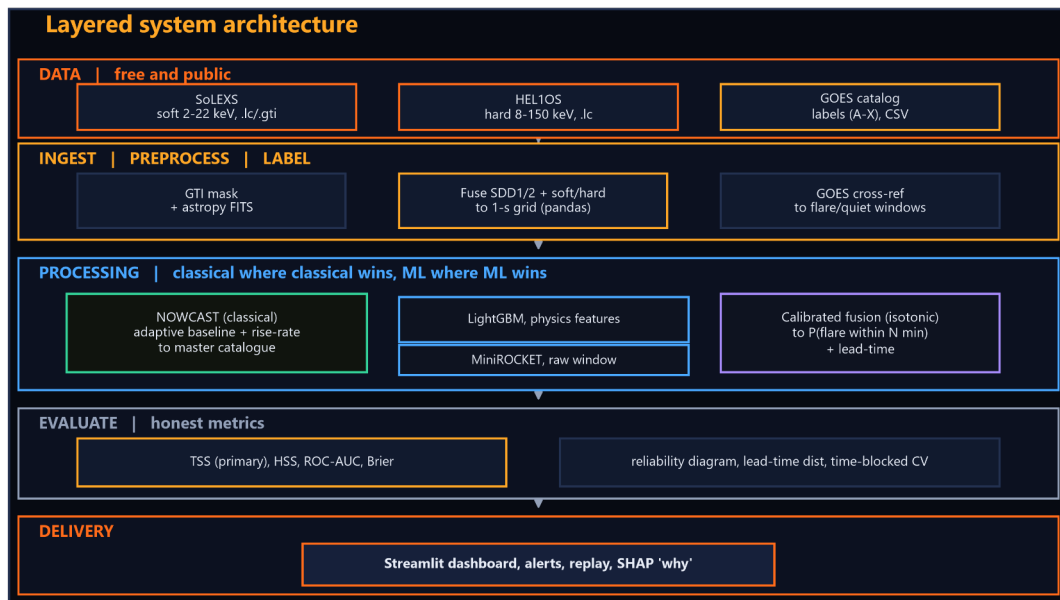
The operator is **put on notice** before anything is certain.

ALERT

A flare is forecast.

The banner shows the predicted **class**, the **minutes of lead time**, and a **calibrated confidence** so the operator can act.

Architecture of the proposed solution

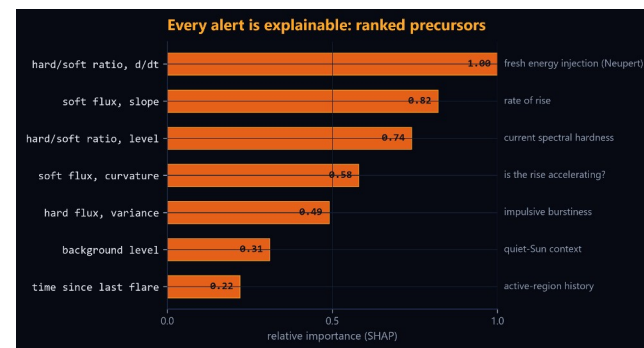
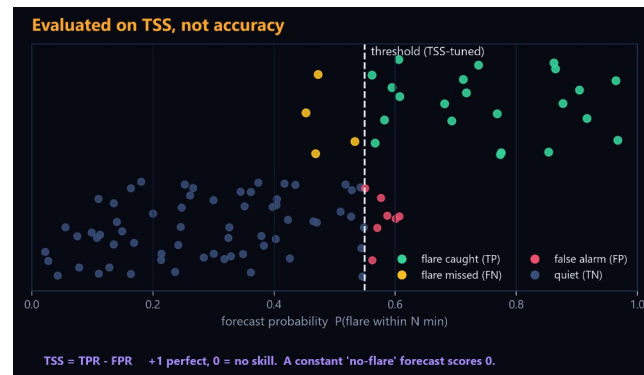


How it works: the cleaned soft and hard light curves feed **two paths at once**.

A **classical signal-processing detector** (adaptive baseline plus rise-rate) handles nowcasting, *interpretable and training-free*.

In parallel, a **LightGBM** gradient-boosted model on Neupert-physics features and a **MiniROCKET** convolutional time-series model on raw windows are fused and **isotonic-calibrated** into one probability, **P(flare within N minutes)**.

Every result is validated with **time-blocked cross-validation** on the **True Skill Statistic**, and **SHAP** attributes each alert to the precursor that drove it.



Technologies used in the solution

Layer	Tool	Why this tool	License
FITS I/O	astropy	standard reader for SoLEXS / HEL1OS products + GTI	BSD
Numerics	numpy, pandas	time-series alignment, resampling, fusion	BSD
Signal	scipy	detrend, smooth, derivative, adaptive baseline	BSD
Forecaster A	LightGBM	fast, interpretable, sample-efficient on rare events	MIT
Forecaster B	MiniROCKET (sktime)	CPU, deterministic, strong on raw time series	BSD
Calibrate, metrics	scikit-learn	isotonic calibration, TSS/HSS/ROC/Brier, blocked CV	BSD
Explainability	SHAP	per-alert precursor attribution (the why)	MIT
Figures	matplotlib, plotly	light curves, reliability, lead-time plots	BSD
Interface	Streamlit	fastest path to dual curves, alerts and replay	Apache
Environment	Python 3.11 (uv)	prebuilt wheels, avoids the 3.14 astropy build break	PSF

Every layer, justified.

The prototype runs on a modest laptop (i5 / RTX 3050 / 16 GB) - proof that the method, not the hardware, does the work.

The same open-source stack scales straight onto ISRO's far larger infrastructure for full-mission throughput.

In ISRO's own tradition: the best results from the least resources.

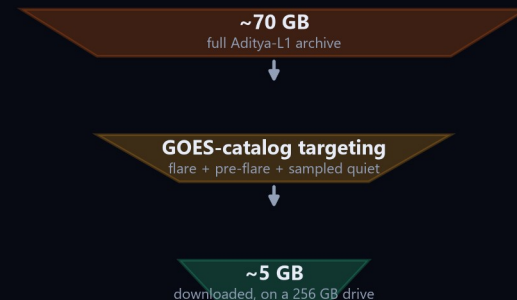
Proof of work and open research: isro-hack-research.vercel.app [Click to Know] - a complete, first-principles companion to the science, data, and methodology behind SuryaSetu,
written and published so any engineer can understand and rebuild the system.

Estimated implementation cost

Prototype (our submission): effectively zero rupees. It needs only an internet connection, a capable GPU (enough VRAM) and a good CPU, all of which we already own.

Item	Cost	Note
Aditya-L1 SoLEXS + HEL1OS data	Rs 0	free, ISSDC PRADAN (registered)
GOES flare catalog (labels)	Rs 0	free, NOAA SWPC
Software stack	Rs 0	all open-source (BSD/MIT/Apache)
Compute	Rs 0	existing laptop, i5 / RTX 3050 / 16 GB
Storage	Rs 0	existing 256 GB drive (~5 GB used)
Hosting (demo)	Rs 0	Streamlit local / free HF Spaces
TOTAL (prototype)	Rs 0	only real input is engineering time

Smart data sampling



We never brute-force 70 GB:
labels point us to the 7% that matters.

Cost at ISRO scale (operational, scoped on selection):

compute at scale (GPU cluster or cloud) | **AI tooling and API keys** | **redundant full-mission storage** | **internet and data egress** | **engineering and MLOps staff.**

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THANK YOU